

Akashdeep Majumder

Software Engineer · Real-Time Systems & Robotics · Frontend & Backend | [Portfolio](#)
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Bangalore, India – 560062

SUMMARY

Software Engineer with **3 years of experience** building real-time systems and Warehouse Management Systems (WMS). Proficient in React, Redux Toolkit, FastAPI, WebSockets, and ROS2. Experienced in Dockerized backend architectures, MySQL database design, 3D visualization using Three.js and React Three Fiber, and state-managed frontend applications. Passionate about scalable system design, performance optimization, and real-time interfaces.

TECHNICAL SKILLS

Frontend: React.js (v18), Redux Toolkit, React Router, React Three Fiber, Three.js, JavaScript (ES6+), Typescript, HTML5, CSS3, Material UI

Database: MySQL

Backend: FastAPI, Django, REST APIs, WebSockets, Redis, Python

DevOps & Tools: Docker, Git, Vite, ElectronJS, CI/CD

Other: WebGL, Immer (undo-redo), MS Office

Robotics: ROS2 (Humble), LiDAR point cloud processing, AMR

WORK EXPERIENCE

Thedush RoboticsMarch 2023 – Present

Software Engineer
Bangalore, India

ONIX – Robot Telemetry & Visualization Software

- Architected a **high-performance 3D visualization pipeline** for Autonomous Mobile Robots (AMRs) using React 18, Three.js, and React Three Fiber (R3F) to track live robot telemetry in real time.
- Optimized LiDAR data streaming (JSON → Float16 + Deflate compression), **reducing bandwidth by ~93% (1.5 MB → 100 KB)** and boosting render performance **from 5 FPS to 50+ FPS**.
- Implemented an **Immer-based undo-redo system** using patch-based state diffing, **cutting memory usage by ~90%** vs. the previous full-snapshot approach.
- Built a real-time dynamic mapping and collision detection system via WebSockets, FastAPI, and ROS2 (Humble).
- Developed a robot task manager for operational control of AMR navigation goals.

Stack: React.js v18, Vite, Three.js, React Three Fiber, Redux Toolkit, WebSockets, Material UI, FastAPI, MySQL, Redis, WebGL

WMS + Robot + PTL Integration System

- Architected and delivered a full **Warehouse Management System (WMS)** integrating Robots and Pick-to-Light (PTL) as a proof of concept for the Indian Army, **in just 14 days**.
- Automated item retrieval with robot navigation + PTL LED guidance, **reducing picking time by ~50%** — store personnel guided directly to items via lit LEDs vs. manual search.
- Replaced manual paper records with a **centralized digital database**, enabling real-time demand/issue tracking and full transaction history.
- Built two distinct UIs (Demand and Issue/Store) and packaged both as standalone desktop apps using Electron.js.

Stack: React.js v18, Vite, Redux Toolkit, WebSockets, ElectronJS, Material UI, FastAPI, MySQL

Robot Configuration Builder

- Designed a React-based guided UI to generate deeply nested **600+ line robot configuration JSON files**, replacing error-prone manual editing for deployment engineers.
- Automated JSON validation and generation, **cutting deployment configuration time by ~50%** and **reducing configuration errors by ~90%**.

Stack: React.js, Vite, Redux Toolkit, Material UI, JavaScript

EDUCATION

Tripura University2019 – 2022

Bachelor of Science in Physics (Honours)